Measuring the Expressive Range of a Procedural Island Generator with Hydraulic Erosion

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Abstract

SeaSail is a real-time C++/OpenGL world of endless ocean and procedurally generated islands. Its terrain is a pure function of position and seed, refined per island by particle-based hydraulic erosion. This report asks what the generator can actually produce and what erosion changes. Using expressive range analysis, I generated 998 islands from 25 seeds and measured each one with and without erosion on a 4 m grid. The four island types occupy distinct regions of the size–height space, with cays and atolls separated by coastline shape rather than size. On 534 mountain islands, erosion lowers mean land height by a median 18% while peaks fall by under 2%, and it raises ruggedness on 81% of islands. Code, data and figures are public and reproducible.

1 Introduction

Procedural content generation (PCG) lets a small team build worlds far larger than they could author by hand [2]. The cost is that nobody sees most of the output. A developer tunes a generator by flying around a few seeds, which says little about the thousands of islands a player may meet. Expressive range analysis addresses this by generating many outputs, measuring each one, and plotting their distribution [1].

This report applies that method to SeaSail, an infinite ocean I built from scratch in C++17 and OpenGL 3.3. It makes three contributions:

1. A deterministic island generator with four island types and cached hydraulic erosion, running in real time with multithreaded streaming (Section 2).
2. A headless measurement tool that samples any island directly from the height function, with no rendering (Section 3).
3. An expressive range analysis of 998 islands, including a paired before-and-after comparison that isolates the effect of erosion (Section 4).

2 System

Island placement. The ocean is divided into 1.5 km cells. A hash of the cell coordinates and the world seed decides whether a cell holds an island and, if so, its type, centre, radius and cliff factor. Centres stay away from cell borders, so any point needs to check only its 9 neighbouring cells. Height is therefore a pure function h(x, z) evaluated in double precision: nothing is stored, the world is unbounded, and any point can be computed on any thread in any order.

Island types. Mountain islands combine fractional Brownian motion for hills with ridged multifractal noise for ridges [3], plus a wandering cliff line on part of the coast. Atolls are a ring of sand around a shallow lagoon, broken by channels. Sea stacks are clusters of steep rock pillars on a shallow shelf. Cays are small, flat sandy islets. A domain warp bends each island's circular outline into bays and headlands, and heights near sea level are compressed so beaches come out wide and flat:

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h' = h - 0.6\, b \tanh(h / b)
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Hydraulic erosion. Mountain islands are eroded by simulated rain droplets, following Beyer's particle model [4]. Each droplet runs downhill with slight inertia, gains speed on steep ground, evaporates, and carries sediment up to a capacity:

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c = \max(-\Delta h, s_{\min}) \cdot v \cdot w \cdot k
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where Δh is the height change of the step, v speed, w water and k a capacity constant. Below capacity it erodes with a soft brush; above, it deposits. Because each droplet changes the ground for the next, erosion cannot be a pure function. It runs once per island on a 384 × 384 grid (about 100,000 droplets, 0.2–0.5 s), and the resulting height change is cached and added to the base terrain. Worker threads that need the same island wait on one shared future, and the cache keeps the 48 most recently used islands.

Runtime. The renderer streams 128 m terrain chunks on worker threads with 6 levels of detail out to about 4.6 km, under a 3 ms per-frame upload budget. Ocean waves are a sum of Gerstner waves [5], evaluated identically on GPU and CPU so the boat floats on the rendered surface.

3 Method

Sample. For each of 25 seeds, every cell in an 8 × 8 block (64 cells, 144 km²) was checked for an island, giving 998 islands: 534 mountain, 178 sea-stack, 159 cay and 127 atoll. Each island was measured twice, from one generator with erosion and one without, so every erosion comparison is paired on the same island.

Sampling. Height was sampled on a 4 m grid over the disc of the island's radius. Points closer, relative to radius, to a neighbouring island's centre were assigned to that neighbour and skipped. The tool calls the same height function the game uses, with no window or GPU; the full run takes about 3.5 minutes on one CPU core.

Metrics. All metrics are computed over land, meaning samples above sea level.

Metric	Definition
Land area	number of land samples × 16 m²
Peak height	maximum land height (m)
Mean slope	mean gradient angle from central differences (°)
Steep share	fraction of land steeper than 35°
Ruggedness	mean absolute height difference to the 8 neighbours (m)
Coast index	coastline length ÷ circumference of a circle of equal area; 1 = round

Coastline length counts land–water cell edges, scaled by π/4 to correct the overestimate of grid edge counting.

4 Results

4.1 The four island types are distinct. Mountains and sea stacks form separate clusters in size and height (Figure 1). Cays and atolls overlap there, but not in shape: their median coast index is 1.02 versus 4.28, because an atoll is a broken ring around a lagoon (Table 1). Mountain islands span the widest range, about 24× in land area and 11× in peak height.

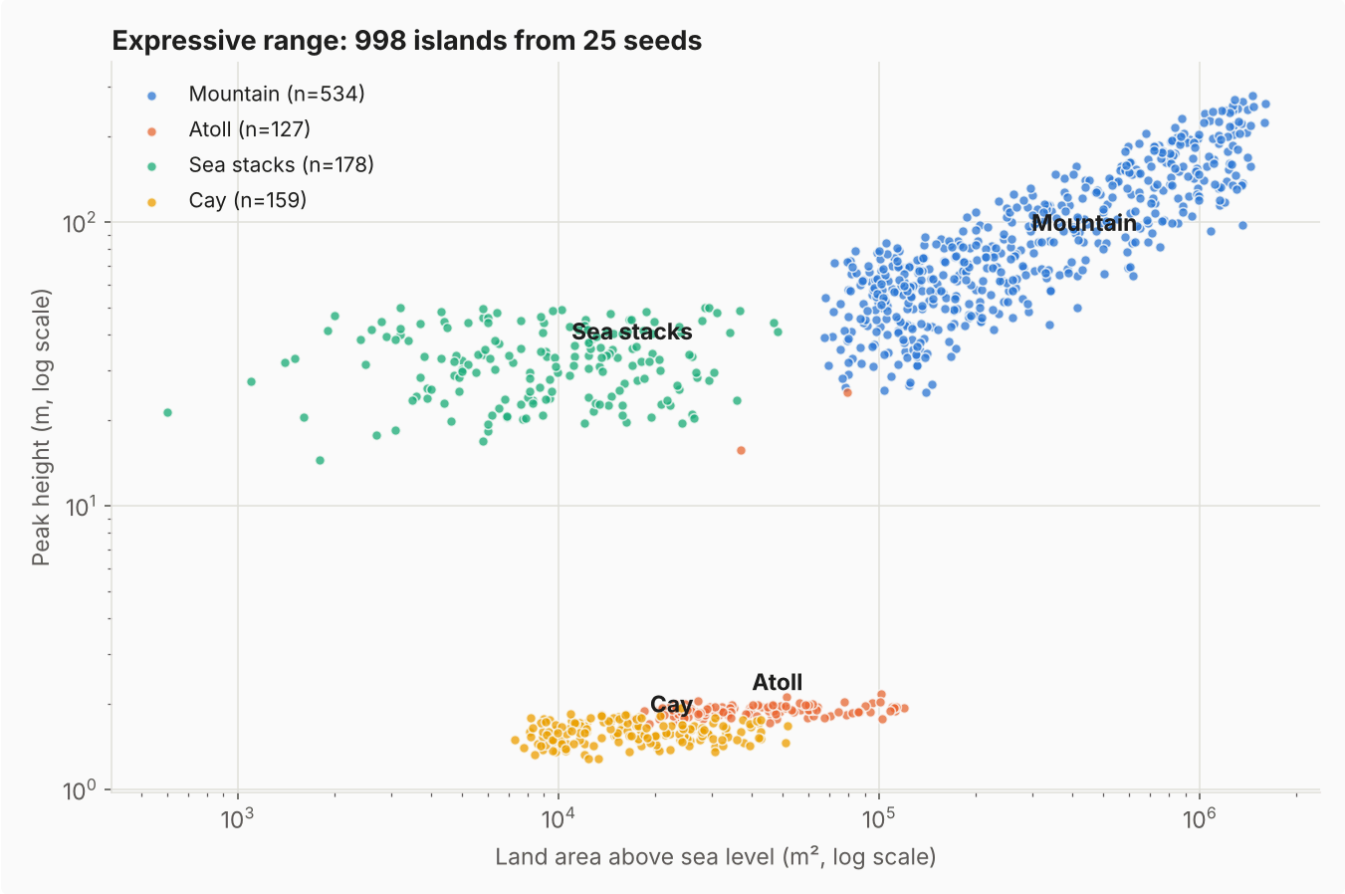


Figure 1. Land area against peak height for all 998 islands, with erosion, by island type.

Type	Islands	Land area (m²)	Peak (m)	Mean slope (°)	Ruggedness (m)	Coast index
Mountain	534	256,950	78.6	23.0	1.55	1.09
Sea stacks	178	9,450	32.5	59.1	9.32	5.39
Cay	159	16,600	1.6	2.4	0.13	1.02
Atoll	127	34,500	1.9	6.7	0.38	4.28

Table 1. Median of each metric by island type, with erosion.

4.2 Mountain islands vary along a steepness–ruggedness axis. Within mountain islands, steepness and ruggedness rise together, but islands with the same share of steep land differ in ruggedness by up to about 1 m (Figure 2). The generator therefore produces more than one kind of mountain island at a given steepness.

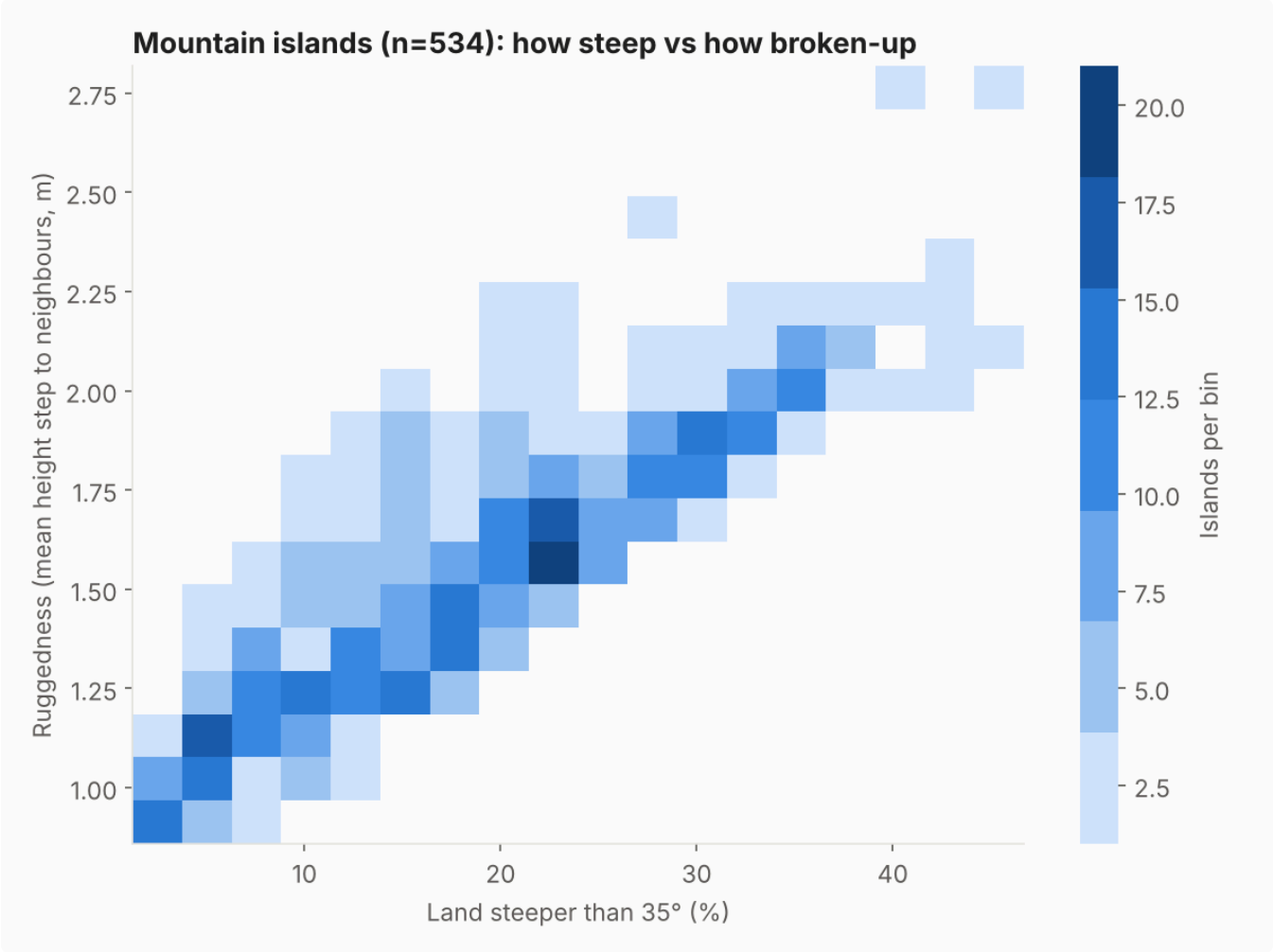


Figure 2. Steep-land share against ruggedness for the 534 mountain islands.

4.3 Erosion carves valleys but keeps the summits. Comparing each mountain island with and without erosion, mean land height falls by a median 18% (middle half: 15–24%) while peak height falls by only 1.8% (Figure 3). Mean land height dropped on every island. Ruggedness rose on 81% of islands (median +9.4%) and steep-land share on 70% (median +16.9%). Land area grew by a median 3.2%, consistent with sediment deposited at the coast. Erosion is also the main cost: sampling one mountain island takes a median 31 ms raw and 263 ms with erosion, including the one-off simulation.

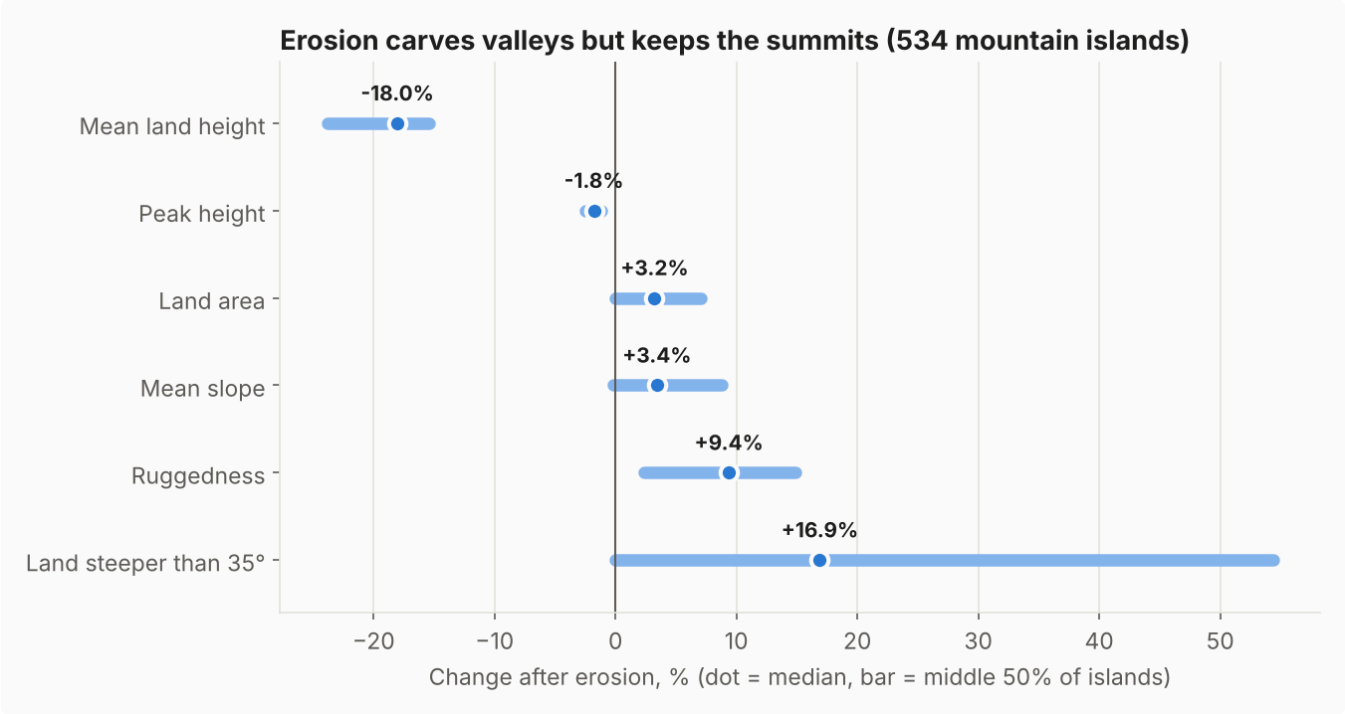


Figure 3. Change after erosion on 534 mountain islands; dot = median, bar = middle 50%.

5 Discussion and limitations

The paired comparison shows erosion behaving as intended: it moves material from slopes into valleys and onto the coast rather than flattening the island. The near-constant peaks matter for gameplay, since summits are the landmarks a sailor navigates by. The wide spread in steep-land change (middle half: 0–54%) suggests erosion's effect depends strongly on the base shape, which is worth modelling directly.

The study has four limits:

- **Neighbour overlap.** Each island is measured on its own disc. Two atolls still pick up land from a neighbouring island, visible as the two atoll points among taller islands in Figure 1.
- **Shape, not experience.** The metrics describe geometry. Whether an island is interesting to explore by boat would need a player study.
- **Fixed parameters.** Erosion settings were held at their defaults, so this report does not show how sensitive the results are to them.
- **One generator.** There is no comparison with other terrain generators, so the ranges are descriptive, not benchmarks.

6 Future work

Three directions would turn this into thesis-scale research:

1. **Controllable erosion.** Sweep the erosion parameters and model how they map onto the measured metrics, so a designer can ask for "more rugged, same peaks" and get it.
2. **Learned island shapes.** Train a model on real coastlines or heightmaps and compare its expressive range with the hand-built generator's, in the spirit of procedural content generation via machine learning [6].
3. **From shape to experience.** Run a player study in SeaSail to test which measured properties predict how interesting an island is to sail around and explore.

References

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Code and data. Source, the measurement tool, raw data (one row per island and erosion setting) and figure scripts: github.com/Nimesh2099/SeaSail. Video: youtu.be/OqiDfUysCDo.